Deterministic Limit of AR(1) Diffusion

Global coupling without innovation noise

Abstract

We remove the AR(1) innovation noise and keep only a random initial state. The recursion becomes

$$a_{k+1} = \alpha a_k, \qquad a_k = \alpha^k A_0, \qquad A_0 \sim \mathcal{N}(\mu_0, \sigma_0^2).$$

The clean trajectory is then supported on the one-dimensional curve vA_0. Here

$$v = (1, \alpha, \dots, \alpha^K)^\mathsf{T}.$$

Its covariance is rank one,

$$\Sigma_0^{\mathrm{det}} = \sigma_0^2 v v^\mathsf{T},$$

so there is no ordinary precision matrix at $t = 0$. After Ornstein–Uhlenbeck corruption,

$$\Sigma_t^{\mathrm{det}} = e^{-2t}\sigma_0^2 v v^\mathsf{T} + (1 - e^{-2t})I,$$

and the inverse can be computed exactly by solving a rank-one linear system. The result is

$$Q_t^{\mathrm{det}} = \frac{1}{\Delta_t} I - \frac{e^{-2t}\sigma_0^2}{\Delta_t\big(\Delta_t + e^{-2t}\sigma_0^2\,\|v\|^2\big)} v v^\mathsf{T}, \qquad \Delta_t = 1 - e^{-2t}.$$

Thus every off-diagonal entry is nonzero for every $t > 0$. The score of each coordinate depends on a single global weighted sum of all coordinates. This shows that loss of locality does not require stochastic innovations: propagation of one shared latent variable already creates global coupling.

1 Setup: removing the innovation noise

Start from the usual AR(1) recursion

$$a_{k+1} = \alpha a_k + \eta_k.$$

The deterministic limit is obtained by setting $\eta_k \equiv 0$, so

$$a_{k+1} = \alpha a_k, \qquad a_k = \alpha^k A_0, \qquad A_0 \sim \mathcal{N}(\mu_0, \sigma_0^2). \tag{1}$$

Collect the trajectory into the vector

$$a = (a_0, a_1, \dots, a_K)^\mathsf{T} \in \mathbb{R}^N, \qquad N = K + 1,$$

and define

$$v := (1, \alpha, \alpha^2, \dots, \alpha^K)^\mathsf{T}. \tag{2}$$

Then the whole trajectory is simply

$$a = vA_0. \tag{3}$$

This is a local recursion in time, but probabilistically it is a single latent variable replicated through the deterministic propagator v.

2 Clean covariance: rank one and globally correlated

The mean is

$$\mu_a := \mathbb{E}[a] = \mu_0 v.$$

Subtracting the mean from (3),

$$a - \mu_a = v(A_0 - \mu_0).$$

Therefore

$$\Sigma_0^{\mathrm{det}} := \mathrm{Cov}(a) = \sigma_0^2 v v^{\mathsf{T}}. \tag{4}$$

Entrywise,

$$(\Sigma_0^{\mathrm{det}})_{ij} = \sigma_0^2 \alpha^{i+j}, \qquad 0 \le i, j \le K. \tag{5}$$

Every pair of coordinates is marginally correlated. The dependence pattern is not a function of distance $|i-j|$, but of the shared latent direction v.

Proposition 2.1 (Rank-one structure)**.** *The clean deterministic covariance has rank one. Its only nonzero eigenvalue is*

$$\lambda_\star = \sigma_0^2 \left\|v\right\|^2, \qquad \left\|v\right\|^2 = \sum_{k=0}^{K} \alpha^{2k} = \begin{cases} \dfrac{1-\alpha^{2N}}{1-\alpha^2}, & \alpha^2 \neq 1, \\ N, & \alpha^2 = 1. \end{cases} \tag{6}$$

Proof. For any vector x,

$$\Sigma_0^{\mathrm{det}} x = \sigma_0^2 v (v^{\mathsf{T}} x).$$

So the image of Σ_0^{det} is the one-dimensional span of v, hence the rank is one. Taking $x = v$ gives

$$\Sigma_0^{\mathrm{det}} v = \sigma_0^2 \left\|v\right\|^2 v,$$

so v is an eigenvector with eigenvalue $\lambda_\star = \sigma_0^2 \left\|v\right\|^2$. Every vector orthogonal to v is sent to zero. $\square$

Because the clean law lives on a one-dimensional subspace, it has no ordinary density on $\mathbb{R}^N$ and therefore no honest precision matrix $Q_0 = (\Sigma_0^{\mathrm{det}})^{-1}$. If one wants an algebraic substitute, the Moore–Penrose pseudoinverse is

$$(\Sigma_0^{\mathrm{det}})^+ = \frac{1}{\sigma_0^2 \left\|v\right\|^4} v v^{\mathsf{T}}, \tag{7}$$

which is already dense. But this pseudoinverse is not a Gaussian conditional-independence object; the clean measure is singular.

3 Adding diffusion noise restores invertibility

Let $z \sim \mathcal{N}(0, I_N)$ be independent of a and define the noisy observation

$$x = e^{-t} a + \sqrt{\Delta_t}\, z, \qquad \Delta_t := 1 - e^{-2t}, \qquad \beta_t := e^{-2t}. \tag{8}$$

Then

$$\mu_t := \mathbb{E}[x] = e^{-t}\mu_a = e^{-t}\mu_0 v,$$

and, using (4),

$$\Sigma_t^{\mathrm{det}} = \beta_t \sigma_0^2 v v^{\mathsf{T}} + \Delta_t I. \tag{9}$$

Now the matrix is strictly positive definite for every $t > 0$, so an ordinary inverse exists.

3.1 Derivation of the inverse from scratch

We solve the linear system

$$(\Delta_t I + \beta_t \sigma_0^2 v v^\mathsf{T}) u = b$$

for arbitrary $b \in \mathbb{R}^N$. Because the perturbation is rank one, it is natural to seek u in the form

$$u = \frac{1}{\Delta_t} b + \lambda v \tag{10}$$

for some scalar λ. Substitute (10) into the equation:

$$\begin{aligned}(\Delta_t I + \beta_t \sigma_0^2 v v^\mathsf{T}) u &= \Delta_t \left(\frac{1}{\Delta_t} b + \lambda v \right) + \beta_t \sigma_0^2 v v^\mathsf{T} \left(\frac{1}{\Delta_t} b + \lambda v \right) \\ &= b + \lambda \Delta_t v + \frac{\beta_t \sigma_0^2}{\Delta_t} v (v^\mathsf{T} b) + \beta_t \sigma_0^2 \lambda \left\| v \right\|^2 v.\end{aligned}$$

For this to equal b, the coefficient of v must vanish:

$$\lambda \Delta_t + \frac{\beta_t \sigma_0^2}{\Delta_t} (v^\mathsf{T} b) + \beta_t \sigma_0^2 \lambda \left\| v \right\|^2 = 0.$$

Hence

$$\lambda = -\frac{\beta_t \sigma_0^2}{\Delta_t \big(\Delta_t + \beta_t \sigma_0^2 \left\| v \right\|^2 \big)} (v^\mathsf{T} b). \tag{11}$$

Substitute (11) back into (10):

$$u = \frac{1}{\Delta_t} b - \frac{\beta_t \sigma_0^2}{\Delta_t \big(\Delta_t + \beta_t \sigma_0^2 \left\| v \right\|^2 \big)} v (v^\mathsf{T} b).$$

Because this holds for every right-hand side b, we have proved the inverse formula.

Theorem 3.1 (Exact deterministic noisy precision)**.** *For every $t > 0$,*

$$Q_t^{\text{det}} := \big(\Sigma_t^{\text{det}} \big)^{-1} = \frac{1}{\Delta_t} I - \frac{\beta_t \sigma_0^2}{\Delta_t \big(\Delta_t + \beta_t \sigma_0^2 \left\| v \right\|^2 \big)} v v^\mathsf{T}. \tag{12}$$

Equivalently, for $i \neq j$,

$$(Q_t^{\text{det}})_{ij} = -\frac{\beta_t \sigma_0^2}{\Delta_t \big(\Delta_t + \beta_t \sigma_0^2 \left\| v \right\|^2 \big)} \alpha^{i+j}. \tag{13}$$

So every off-diagonal entry is nonzero whenever $\alpha \neq 0$.

The deterministic benchmark therefore has a very different dense pattern from the Gaussian AR(1) benchmark. In the Gaussian model, the long-range entries are created by inverting a matrix with local structure and typically decay with distance $|i - j|$. In the deterministic model, the off-diagonal part is the separable rank-one matrix $v v^\mathsf{T}$.

4 Score: every coordinate sees a global weighted sum

Because x is Gaussian for every $t > 0$, the score is affine:

$$S^{\det}(x,t) = \nabla_x \log p_t(x) = -Q_t^{\det}(x - \mu_t). \tag{14}$$

Insert (12). If we define

$$c_t := \frac{\beta_t \sigma_0^2}{\Delta_t\big(\Delta_t + \beta_t \sigma_0^2 \,\|v\|^2\big)}, \tag{15}$$

then

$$S^{\det}(x,t) = -\frac{x - \mu_t}{\Delta_t} + c_t\, v\, v^{\mathsf{T}}(x - \mu_t). \tag{16}$$

The kth component is therefore

$$S_k^{\det}(x,t) = -\frac{x_k - \mu_{t,k}}{\Delta_t} + c_t \alpha^k \sum_{j=0}^{K} \alpha^j (x_j - \mu_{t,j}). \tag{17}$$

This is the key structural fact: *every score component depends on the same global scalar statistic*

$$\sum_{j=0}^{K} \alpha^j (x_j - \mu_{t,j}).$$

The coupling is global even though the underlying recursion (1) is one-step local.

5 Comparison with the stochastic Gaussian AR(1) model

5.1 What disappears when $\sigma_\eta^2 \to 0$?

In the stochastic Gaussian AR(1) model,

$$a_{k+1} = \alpha a_k + \eta_k, \qquad \eta_k \sim \mathcal{N}(0, \sigma_\eta^2),$$

the clean precision is tridiagonal:

$$Q_0^{\mathrm{sto}} = \begin{pmatrix} \sigma_0^{-2} + \alpha^2 \sigma_\eta^{-2} & -\alpha\sigma_\eta^{-2} & & \\ -\alpha\sigma_\eta^{-2} & (1+\alpha^2)\sigma_\eta^{-2} & \ddots & \\ & \ddots & \ddots & -\alpha\sigma_\eta^{-2} \\ & & -\alpha\sigma_\eta^{-2} & \sigma_\eta^{-2} \end{pmatrix}.$$

As $\sigma_\eta^2 \downarrow 0$, these entries blow up: the quadratic form enforces the hard constraints $a_{k+1} = \alpha a_k$. In covariance language the full-rank Gaussian law collapses onto the rank-one deterministic manifold.

For any fixed $t > 0$, however, the noisy covariance stays invertible:

$$\Sigma_t^{\mathrm{sto}}(\sigma_\eta^2) = e^{-2t}\Sigma_0^{\mathrm{sto}}(\sigma_\eta^2) + \Delta_t I.$$

Since $\Sigma_0^{\mathrm{sto}}(\sigma_\eta^2) \to \sigma_0^2 v v^{\mathsf{T}}$ entrywise as $\sigma_\eta^2 \to 0$, we immediately obtain

$$\Sigma_t^{\mathrm{sto}}(\sigma_\eta^2) \to \Sigma_t^{\det}, \qquad Q_t^{\mathrm{sto}}(\sigma_\eta^2) \to Q_t^{\det}. \tag{18}$$

So the deterministic benchmark is not a separate curiosity; it is the regularized small-noise limit of the Gaussian AR(1) model.

5.2 What is the conceptual difference?

The stochastic Gaussian benchmark and the deterministic benchmark separate two notions that are easy to confuse:

1. **Local dynamics.** Both models are generated by a one-step recursion.
2. **Probabilistic Markov locality.** Only the full-rank Gaussian model has a clean tridiagonal precision encoding conditional independence.
3. **Global latent coupling.** In the deterministic limit all coordinates are functions of the same scalar A_0, so diffusion reveals a dense rank-one interaction pattern.

Thus randomness is not what creates global coupling. Propagation through a shared latent variable already does it.

6 Numerical illustrations

All figures were generated by the reproducible script `code/generate_figures.py`. We used $\alpha = 0.9$, $\sigma_0^2 = 1$, and $N = 60$.

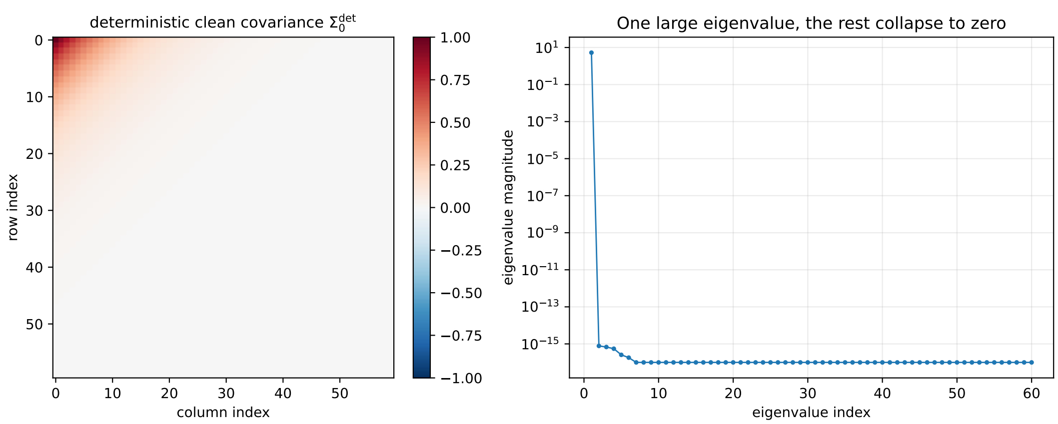

Figure 1: Deterministic clean covariance. Left: heatmap of $\Sigma_0^{\text{det}} = \sigma_0^2 v v^\top$, which is dense although rank one. Right: eigenvalues on a log scale, showing one nonzero mode and numerical zeros for the rest.

The figures support four concrete statements:

1. The clean deterministic covariance is globally correlated even before diffusion.
2. The regularized precision for $t > 0$ is dense with a rank-one off-diagonal pattern.
3. This density does not come from stochastic innovations; it comes from the shared latent coordinate A_0.
4. The deterministic benchmark is the small-noise limit of the Gaussian AR(1) benchmark at any fixed $t > 0$.

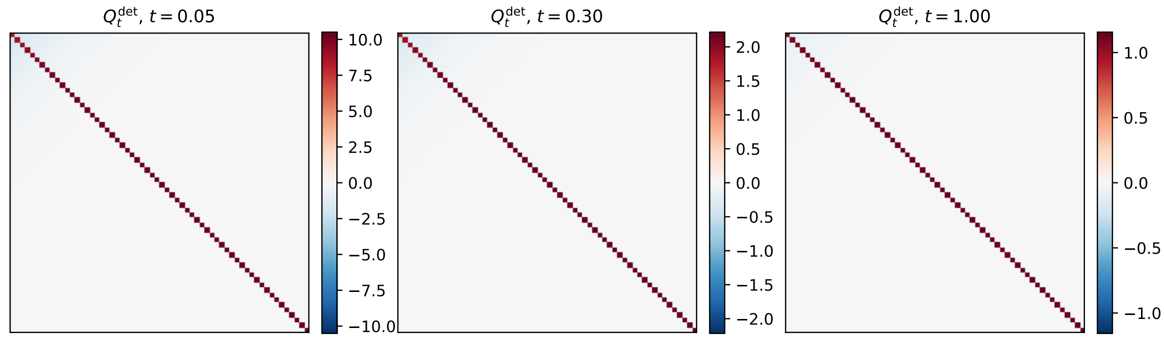

Figure 2: Heatmaps of Q_t^{det} at three diffusion times. The matrix is always diagonal minus a dense rank-one correction. No notion of nearest-neighbor sparsity survives.

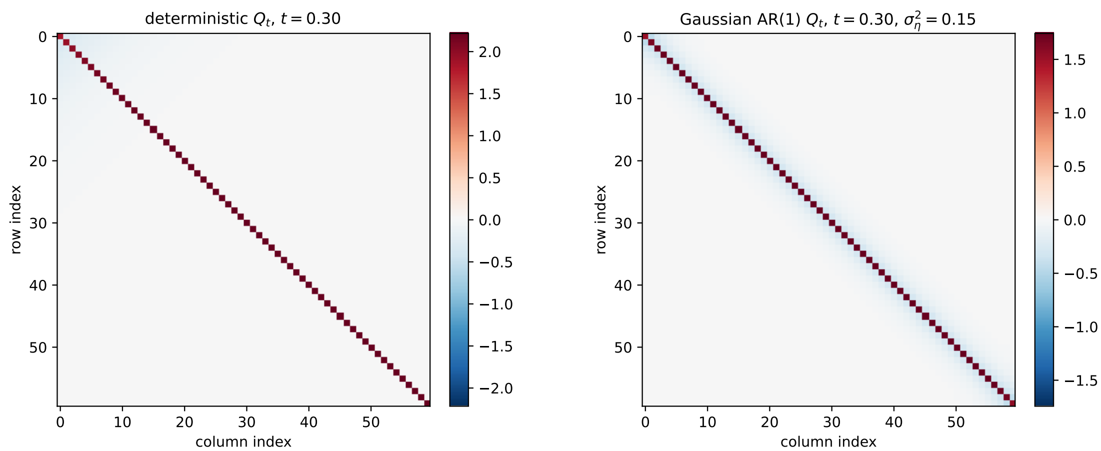

Figure 3: Deterministic versus stochastic Gaussian AR(1) at the same diffusion time. The deterministic pattern is globally dense and strongly aligned with the latent direction v. The stochastic Gaussian pattern reflects Markov fill-in from an originally tridiagonal precision.

7 Takeaway

The deterministic limit isolates propagation from randomness. Once the innovations are removed, the trajectory becomes

$$(a_0, a_1, \ldots, a_K) = A_0(1, \alpha, \ldots, \alpha^K),$$

so all coordinates inherit the same latent scalar. The clean covariance collapses to rank one and ceases to admit a full-dimensional precision. After any positive diffusion time, the covariance becomes invertible and its inverse is explicit:

$$Q_t^{\mathrm{det}} = \frac{1}{\Delta_t} I - c_t v v^\mathsf{T}.$$

This matrix is dense, not because of Markov noise, but because of deterministic propagation through a shared latent mode. In this sense, the deterministic study cleanly separates two effects:

$$\text{Markovianity} \neq \text{global coupling}.$$

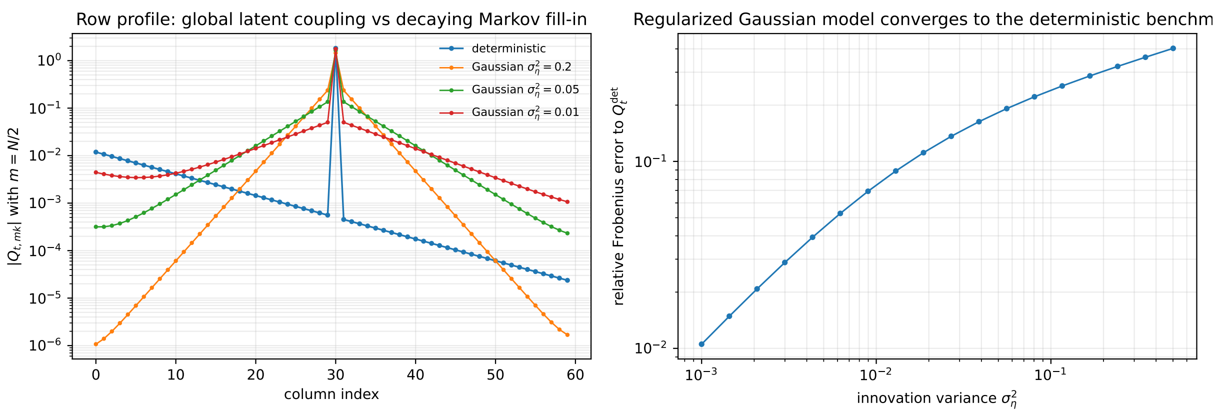

Figure 4: Left: a row profile of the precision. The deterministic benchmark couples one row to all columns through the shared latent direction. The Gaussian model has a different profile, but approaches the deterministic one as the innovation variance decreases. Right: the regularized Gaussian precision converges to Q_t^{det} as $\sigma_\eta^2 \to 0$, confirming (18).

The Gaussian AR(1) model has local conditional structure. The deterministic limit has global latent coupling. Diffusion regularizes both, but the two mechanisms are different.